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$$w = \ln(x^2 + y^2 + z^2)$$

$$x = s - t$$

$$y = s + t$$

$$z = 2\sqrt{st}$$

$$\begin{aligned}\frac{\partial w}{\partial s} &= \frac{\partial w}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial w}{\partial y} \cdot \frac{\partial y}{\partial s} + \frac{\partial w}{\partial z} \cdot \frac{\partial z}{\partial s} \\ &= \frac{2x}{x^2 + y^2 + z^2} + \frac{2y}{x^2 + y^2 + z^2} + \frac{2z}{x^2 + y^2 + z^2} \cdot \frac{t}{\sqrt{st}} \\ &= \frac{2}{s+t} \\ \frac{\partial w}{\partial t} &= \frac{\partial w}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial w}{\partial y} \cdot \frac{\partial y}{\partial t} + \frac{\partial w}{\partial z} \cdot \frac{\partial z}{\partial t} \\ &= -\frac{2x}{x^2 + y^2 + z^2} + \frac{2y}{x^2 + y^2 + z^2} + \frac{2z}{x^2 + y^2 + z^2} \cdot \frac{s}{\sqrt{st}} \\ &= \frac{2}{s+t}\end{aligned}$$

References